



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2021
General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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CHEMISTRY

Paper 2 Structured Questions

8873/02

16 September 2021

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not use paper clips, highlighters, glue or correction fluid.

Write your name, civics group, registration number on all the work you hand in.

Write in **dark blue or black pen**.

You may use an HB pencil for any diagrams or graphs.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

For Examiner's Use	
Section A	
1	/ 14
2	/ 8
3	/ 15
4	/ 23
Section B	
5 / 6	/ 20
Total	/ 80

This document consists of **23** printed pages and **1** blank page.

Section A

Answer all questions in this section.

- 1 The fourth ionisation energies of the elements **A** to **G**, proton number ranging from 6 to 20, are shown in Fig. 1.1.

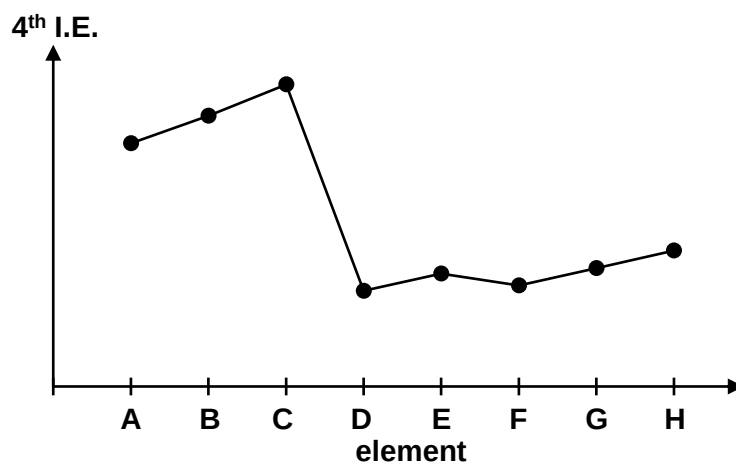


Fig. 1.1

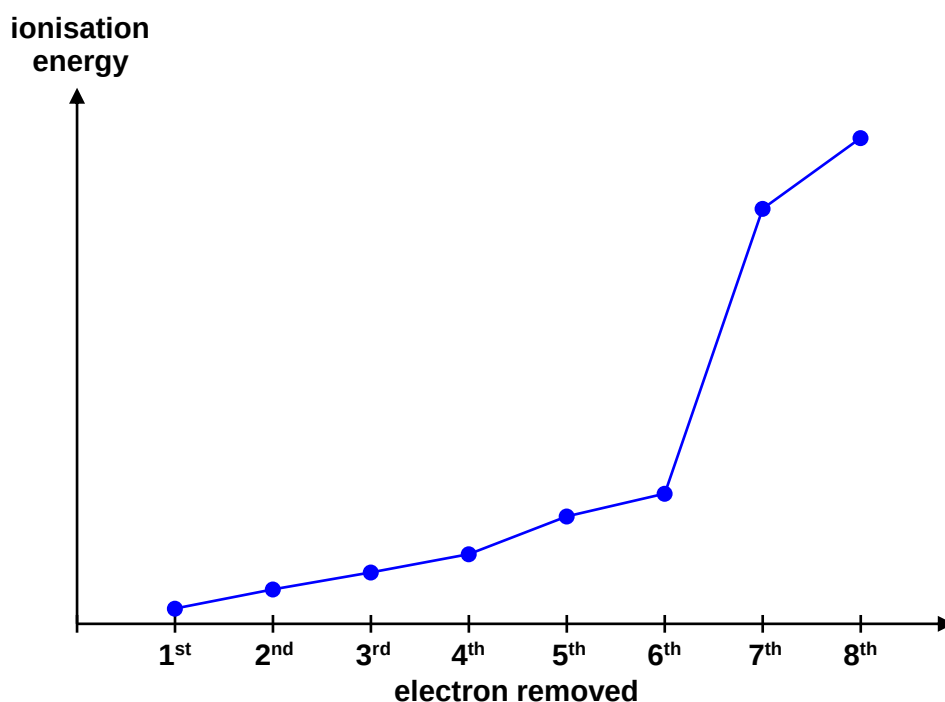
- (a) (i) Explain why element **F** is sulfur.

From the diagram, there is a drastic drop in 4th IE from element C to D,
showing that the 4th electron removed from D is from the valence shell while
that of C is from an inner quantum shell, thus element **D** is from Group 14.
 Since the elements range from proton number 6 to 20, hence element A is
from Group 1 and element F is from Group 16 of the same period, element
A can only be sodium and element F is sulfur. [2]

- (ii) Write the electronic configuration of **F³⁺** and **E³⁺**. Hence, explain why element **F** has a lower 4th ionisation energy than element **E**.

1s² 2s² 2p⁶ 3s² 3p¹ E³⁺: 1s² 2s² 2p⁶ 3s²
 The 3p electron to be removed from **F³⁺** is at a higher energy level than the
2s electron to be removed from **E³⁺**. Thus, less energy is required to remove
the 2p electron, resulting in a lower ionisation energy. [2]

(iii) Sketch the first 8 successive ionisation energies of element **F** in the axes below.



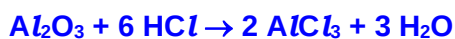
[1]

(b) Oxides of Period 3 elements exhibit acid/base behaviour. With the aid of suitable equations, describe the nature of sodium oxide, aluminium oxide and phosphorous pentoxide when reacted with hydrochloric acid and/or sodium hydroxide.

Na_2O is basic in nature and reacts with acid to form salt and water.



Al_2O_3 is amphoteric and can react with both acids and bases.



P_4O_{10} is acidic in nature and reacts with base to form salt and water...



[3]

- (c) Magnesium chloride, silicon tetrachloride and phosphorus pentachloride are some of the chlorides of Period 3 elements. With reference to structure and bonding, rank the melting points of these three chlorides in decreasing order. Explain your answer.

Melting point: $\text{MgCl}_2 > \text{PCl}_5 > \text{SiCl}_4$

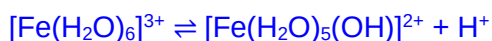
MgCl_2 has the highest melting point as it has a giant ionic structure whereby Mg^{2+} and Cl^- are held together by strong electrostatic forces of attraction. Thus, a large amount of energy is required to overcome these strong ionic bonds.

Both PCl_5 and SiCl_4 are simple molecules held together by weak intermolecular instantaneous dipole – induced dipole (id-id) attraction, thus have lower melting points than MgCl_2 .

However, PCl_5 has a larger electron cloud than SiCl_4 , thus the electron cloud of PCl_5 is more polarisable, resulting in more energy required to overcome the stronger intermolecular id-id and thus a higher melting point. [3]

- (d) When aluminium chloride is added to water with universal indicator, the solution turned orange. A similar observation was made when iron(III) chloride was added to water. Using appropriate values from the *Data Booklet*, suggest a reason and write the equation(s) of the reaction of iron(III) chloride with water.

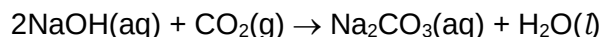
Both Al^{3+} and Fe^{3+} have the same ionic charge and similar ionic radii (Al^{3+} : 0.050 nm, Fe^{3+} : 0.055 nm), thus they have similar charge densities. Fe^{3+} is thus able to polarise the electron cloud of surrounding water molecules, weakening and breaking the O-H bond, resulting in a release of a proton to give an acidic solution.



[3]

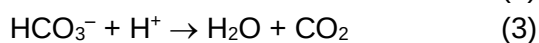
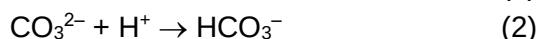
[Total: 14]

- 2 4.50 g of a monobasic organic acid, $C_xH_{2x+1}CO_2H$, is burnt completely in excess of oxygen. The carbon dioxide gas evolved is completely absorbed into 1 dm³ of sodium hydroxide solution:



To determine the concentration of sodium hydroxide used, the resultant solution was titrated against hydrochloric acid solution.

During the titration, the reaction between sodium hydroxide and hydrochloric acid occurs in a single step, while that between sodium carbonate and hydrochloric acid occurs in two steps with hydrogencarbonate ion as the intermediate product.



With phenolphthalein as indicator, the end-point is seen at the completion of reactions (1) and (2). With methyl orange as indicator, the end-point is seen at the completion of all three reactions.

- (a) Suggest a reason why excess sodium hydroxide is used to absorb carbon dioxide gas.

To ensure that all $CO_2(g)$ has been fully reacted away...... [1]

- (b) From the 1 dm³ resultant solution, 10 cm³ samples were titrated against 0.250 mol dm⁻³ hydrochloric acid. When one sample was titrated in the presence of phenolphthalein, 16.00 cm³ of hydrochloric acid was required to reach the end-point. When a second sample was titrated in the presence of methyl orange, 24.20 cm³ of hydrochloric acid was required to reach the end-point.
- (i) Calculate the amount of hydrochloric acid that has reacted in both samples. Hence, show that in a 10 cm³ sample, the amount of hydrogencarbonate ion present is 0.00205 mol.

$$\text{Using phenolphthalein: amt of HCl} = \frac{16.00}{1000} \times 0.25 = 0.00400 \text{ mol}$$

$$\text{Using methyl orange: amt of HCl} = \frac{24.20}{1000} \times 0.25 = 0.00605 \text{ mol}$$

$$n(HCl_{\text{phenolphthalein}}) = n(NaOH_{\text{rxn 1}}) + n(Na_2CO_{3, \text{rxn 2}})$$

$$n(HCl_{\text{methyl orange}}) = n(NaOH_{\text{rxn 1}}) + n(Na_2CO_{3, \text{rxn 2}}) + n(NaHCO_{3, \text{rxn 3}}) \text{ thus}$$

$$n(HCl_{\text{methyl orange}}) - n(HCl_{\text{phenolphthalein}}) = n(NaHCO_{3, \text{rxn 3}})$$

$$\begin{aligned} \text{Thus, difference in amt of HCl} &= \text{amt of } HCO_3^- \text{ reacted in rxn 3} \\ &= 0.00605 - 0.00400 = 0.00205 \text{ mol} \end{aligned}$$

[2]

- (ii) Using the value given in (b)(i), calculate the respective amount of carbonate ions and hydroxide ions present in the resultant solution.

$$\text{amt of HCO}_3^- \text{ from (i)} = \text{amt of CO}_3^{2-} \text{ in } 10 \text{ cm}^3 \text{ sample}$$

$$\text{amt of CO}_3^{2-} \text{ in } 1 \text{ dm}^3 = 0.00205 \times \frac{1000}{10} = 0.205 \text{ mol}$$

$$\begin{aligned} \text{amt of OH}^- \text{ in } 10 \text{ cm}^3 &= 0.00400 - 0.00205 \text{ mol} \\ &= 0.00195 \text{ mol} \end{aligned}$$

OR

$$\begin{aligned} \text{amt of OH}^- \text{ in } 10 \text{ cm}^3 &= 0.00605 - 2(0.00205) \text{ mol} \\ &= 0.00195 \text{ mol} \end{aligned}$$

$$\text{amt of OH}^- \text{ in } 1 \text{ dm}^3 = 0.195 \text{ mol}$$

[2]

- (iii) Hence, calculate the concentration of sodium hydroxide used to absorb carbon dioxide gas.

$$\begin{aligned} \text{Total amt of OH}^- &= 2 \times \text{amt of CO}_3^{2-} + \text{amt of OH}^- \text{ unreacted} \\ &= 2 \times 0.205 + 0.195 = 0.605 \text{ mol} \end{aligned}$$

$$[\text{NaOH}] = 0.605 \text{ mol dm}^{-3}$$

[1]

- (iv) Using the value given in (b)(i), deduce the value of x in the organic acid, $\text{C}_x\text{H}_{2x+1}\text{COOH}$.

$$\text{General formula of a saturated monobasic acid: } \text{C}_x\text{H}_{2x+1}\text{CO}_2\text{H} = \text{C}_{x+1}\text{H}_{2x+2}\text{O}_2$$

$$\text{amt of CO}_3^{2-} = \text{amt of CO}_2 \text{ absorbed} = 0.205 \text{ mol}$$

$$\text{amt of organic acid} = \frac{\text{amt of CO}_2}{x+1} \text{ since } \text{C}_x\text{H}_{2x+1}\text{CO}_2\text{H} \equiv (x+1)\text{CO}_2$$

$$\frac{4.5}{12(x+1)+1(2x+2)+16(2)} = \frac{0.205}{x+1}$$

$$4.5(x+1) = 0.205(14x + 46)$$

$$4.5x + 4.5 = 2.87x + 9.43$$

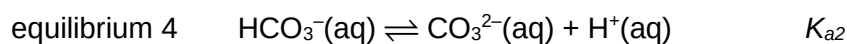
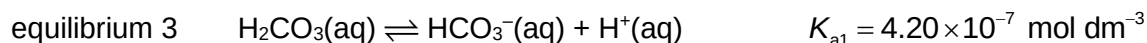
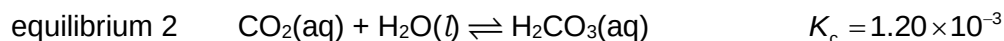
$$x = 3.02 = \underline{3}$$

Thus, the molecular formula is $\text{C}_4\text{H}_8\text{O}_2$

[2]

[Total: 8]

- 3 When carbon dioxide from the atmosphere dissolve in the ocean, it forms carbonic acid, H_2CO_3 , which in itself is unstable. This further dissociates to form hydrogencarbonate and carbonate ions.



where K_{a2} is very small.

- (a) With reference to structure and bonding, suggest why $\text{CO}_2(\text{g})$ dissolves better in water than $\text{O}_2(\text{g})$ and $\text{N}_2(\text{g})$.

While all gases are simple, non-polar molecules, CO_2 contains polar C=O bonds

that allow it to form stronger interaction with water.

[2]

- (b) (i) Write the K_{a1} expression for equilibrium 3.

$$K_{a1} = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3]}$$

[1]

- (ii) A sample of ocean water has a pH of 8.1. What is the concentration of H^+ ions in ocean water?

$$[\text{H}^+] = 10^{-8.1} = 7.9433 \times 10^{-9} = 7.94 \times 10^{-9} \text{ mol dm}^{-3}$$

[1]

- (iii) Assuming the concentration of H^+ in ocean water is mainly from equilibrium 3, use your answer in (b)(ii) to calculate the equilibrium concentration of $\text{H}_2\text{CO}_3(\text{aq})$ in equilibrium 3.

$$K_{a1} = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3]}$$

$$= \frac{(7.94 \times 10^{-9})(7.94 \times 10^{-9})}{[\text{H}_2\text{CO}_3]}$$

$$[\text{H}_2\text{CO}_3] = \frac{(7.94 \times 10^{-9})(7.94 \times 10^{-9})}{(4.20 \times 10^{-7})}$$

$$= 1.3424 \times 10^{-10} = 1.34 \times 10^{-10} \text{ mol dm}^{-3}$$

[2]

(iv) Hence, calculate the equilibrium concentration of dissolved CO₂ in equilibrium 2.

$$K_c = \frac{[\text{H}_2\text{CO}_3]}{[\text{CO}_2(\text{aq})]}$$

$$[\text{CO}_2(\text{aq})] = \frac{(1.3424 \times 10^{-10})}{(1.20 \times 10^{-3})} = 1.118 \times 10^{-7} = 1.12 \times 10^{-7} \text{ mol dm}^{-3}$$

[1]

- (c) Since the Industrial Revolution, human activity has increased the level of atmospheric CO₂, which in turn lowered the pH of the ocean. Using equilibrium 1 to 4, explain this phenomenon.

More carbon dioxide gas present in the atmosphere. Thus, by Le Chatelier's

Principle, position of equilibrium 1 shifts to the right to counteract the increase

in amount of CO₂(g).

This causes the concentration of CO₂(aq) to increase, causing the position of

equilibrium 2 to shift right. This causes the equilibrium concentration of

H₂CO₃(aq) to increase, which would cause position of equilibrium 3 and in turn

equilibrium 4 to shift right. This results in the concentration of H⁺(aq) to

increase.

Since pH = -lg[H⁺], the pH of the ocean decreases.

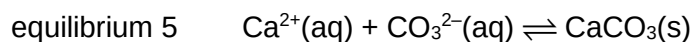
[4]

- (d) The HCO₃⁻ / CO₃²⁻ system acts as a buffer for the ocean. State the equation showing how the buffer responds to the lowering of pH in the ocean.

The increase in H⁺ will be removed by CO₃²⁻: CO₃²⁻ + H⁺ → HCO₃⁻

[1]

Marine organisms such as mussels and corals use the abundant calcium ions present in the ocean to form CaCO_3 shells and exoskeletons.



- (e) (i) Using your answer in (d) and equilibrium 5, suggest the impact of sustained increased level of atmospheric CO_2 on marine organisms such as corals and mussels.

More CO_3^{2-} will be consumed and affecting the growth of mussels and corals/ corrode exoskeleton of corals.

..... [1]

- (ii) Explain if equilibrium 5 is exothermic or endothermic, given that $\text{CaCO}_3(\text{s})$ dissolves less easily with rising temperature.

By Le Chatelier's Principle, increase in temperature favours endothermic reaction to absorb heat. CaCO_3 dissolve less easily meant the forward reaction of eqm 5 is favoured. Hence, eqm 5 is endothermic.

..... [2]

[Total: 15]

- 4 Poly(ethene) is the simplest addition polymer containing only singly bonded carbon and hydrogen atoms. It is durable, tough and have excellent electrical insulating properties.

Ethene, CH_2CH_2 , forms poly(ethene) of different molecular structures by varying temperature, pressure and catalyst used.

Table 4.1 shows some data for poly(ethene) formed under different conditions.

Table 4.1

	density / g cm^{-3}	tensile strength / MPa	elongation at fracture / %	uses
A	0.92	15	600	plastic wrap
B	0.97	29	350	milk bottles
cross-linked poly(ethene) (CLPE)	0.93	18	270	bottle caps

Tensile strength is the amount of load or stress that can be handled by a material before it stretches and breaks.

Elongation at fracture is the ratio between changed length and initial length after breakage of the test specimen.

- (a) (i) Explain what is meant by the term *polymer*.

Polymers are macromolecules built up from monomers, with average molar mass of at least 1000 or at least 100 repeat units. [1]

- (ii) Briefly describe in terms of bonding, what happens during the addition polymerisation of ethene.

The π bond about $\text{C}=\text{C}$ breaks homolytically forming new $\text{C}-\text{C}$ bond with other monomers. [1]

- (iii) Explain the change in bond angle about carbon atom from ethene to poly(ethene), using VSEPR theory.

In ethene, there are 3 bond pair and 0 lone pair of electrons about carbon while there are 4 bond pairs and no lone pair about carbon in poly(ethene). Hence to minimise repulsion, the bond angle decreases from 120° to 109.5°

[2]

- (iv) The formation of poly(ethene) is catalysed by solid chromium oxide, a heterogeneous catalyst. Outline the mode of action for this type of catalysis.

Chromium oxide provides active sites whereby ethene molecules are adsorbed onto the catalyst surface via weak bonds. This increases the surface concentration of the ethene molecules and weakens the C=C bond, thus lowering the activation energy for reaction to occur. The adsorbed product molecules then breaks free from the catalyst surface via desorption.

..... [3]

- (b) Suggest if any of the polymers in Table 4.1 will sink in water.

All the polymers in Table 4.1 will float/ not sink in water as their density are lower than that of water (1 g cm^{-3}).

[1]

- (c) Using information from Table 4.1 and your knowledge of the structure and bonding in polymers, answer the following questions:

- (i) A and B could be either LDPE or HDPE. Deduce, with reasoning, the identity of A and B.

A is LDPE. B is HDPE.

A has lower tensile strength/higher elongation as LDPE is a branched-chain polymer which there are branches extending from the backbone of a polymeric chain. The branches of LDPE prevent maximum surface area of contact between the polymeric molecules. Or Chains cannot be packed closely together, creating space between the polymer chains. As a result, the instantaneous dipole-induced dipole (i.d.-i.d.) interactions between the molecules are not extensive / weaker leading to lower tensile strength. The polymer chains are also able to slide past each other easily.

B has higher tensile strength/lower elongation as HDPE is linear polymer which are able to stack closer/have higher surface area of contact.

[4]

This results in stronger id-id between the HDPE polymer chains.

- (ii) With reference to your answer in (c)(i), state a relevant physical property that allow **A** and **B** to be used according to table 4.1.

LDPE has weaker intermolecular interactions than HPDE. Hence **A** is able to stretch/more flexible / elongation factor to be used as a plastic wrap while **B** has higher melting point and is able to withstand heat or more durable/ able to hold its shape / higher tensile strength as a material for milk bottle.

..... [2]

- (d) Among the three types of poly(ethene), CLPE is unsuitable for recycling or remoulding. Suggest, with reasoning whether CLPE is a thermoplastic or thermoset polymer.

CLPE is a thermoset polymer. The cross links are strong covalent bonds connecting the polymer chains. Hence they are able to retain their rigidity under high temperature.

..... [2]

- (e) The durable properties of poly(ethene) also posed environmental issues.

- (i) Suggest why poly(ethene) is non-biodegradable.

The polymer is made of non polar C-C and C-H bonds that does not react easily.

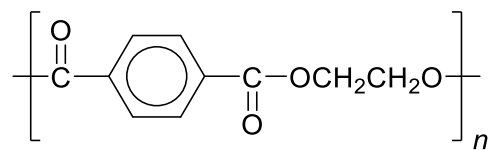
..... [1]

- (ii) Carbonyl groups (C=O) are incorporated into poly(ethene) to allow it to become photodegradable by absorbing UV light. Suggest one situation where the use of photodegradable polymer is unsuitable.

Use in outdoor products e.g. traffic cones, car dashboard components etc

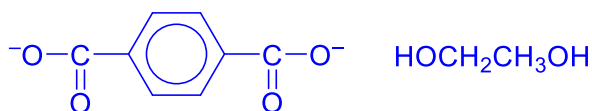
..... [1]

- (f) Poly(ethylene terephthalate) are also used to make bottles.



poly(ethylene terephthalate) or PET

PET bottles are not suitable to store strongly alkaline solutions. State the reaction that may occur between the PET bottle and the alkaline solution and draw the structure of products formed.

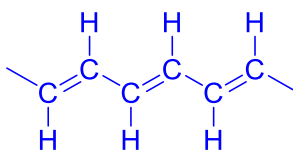


Alkaline hydrolysis. [2]

- (g) Ethyne, the simplest alkyne, has the structure $\text{HC}\equiv\text{CH}$. Similar to ethene, ethyne undergoes addition polymerisation to form poly(ethyne).

The discovery and development of poly(ethyne) as an organic conducting polymer led to Nobel Prize in Chemistry in 2020. The electrical conductivity of poly(ethyne) is increased when small amount of iodine is added. In this reaction, iodine acts as an oxidising agent.

- (i) Draw three repeating units of poly(ethyne).



[1]

- (ii) With reference to your answer in (g)(i), account for the electrical conductivity of poly(ethyne) chain.

The π bonding electrons act as mobile charge carriers as they are delocalised due to the alternate single and double bonds across the polymer chain. [1]

- (iii) Suggest the effect of adding bromine instead of iodine to the electrical conductivity of poly(ethyne).

Br_2 is acting as a stronger oxidising agent as it is more likely to gain electron due to stronger effective nuclear charge. Hence it can increase the electrical conductivity more than that of iodine. [1]

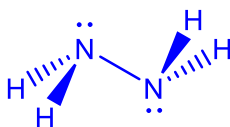
[Total: 23]

Section B

Answer **one** of the questions in this section.

- 5 (a) Hydrazine is an inorganic compound with the chemical formula, N_2H_4 . It is a colourless flammable liquid with an ammonia-like odour.

- (i) Draw the structure of hydrazine, stating the shape and bond angle about each central atom, N.



Shape about N: **Trigonal pyramidal**

Bond angle: **107°**

[3]

- (ii) Hydrazine can be synthesised via the following reaction under aqueous conditions.



State and explain the type of reaction shown above.

Redox reaction

N is oxidised from -3 in NH_3 to -2 in N_2H_4 while Cl is reduced from +1 in ClO^-

to -1 in Cl^- [2]

- (iii) A series of experiments was carried out at different concentrations of ClO^- and NH_3 .

Table 5.1 shows the results obtained.

Table 5.1

experiment	$[\text{ClO}^-] / \text{mol dm}^{-3}$	$[\text{NH}_3] / \text{mol dm}^{-3}$	initial rate $/ \text{mol dm}^{-3} \text{ s}^{-1}$
1	0.200	0.100	0.256
2	0.400	0.200	2.05
3	0.400	0.400	8.2

Using the data in Table 5.1, determine the order of reaction with respect to each reacting species and the rate equation of this reaction.

Comparing Expt 2 and 3,

When $[\text{NH}_3]$ doubled, $[\text{ClO}^-]$ is held constant, rate increased 4 times. Thus order of reaction with respect to NH_3 is 2.

Comparing Expt 1 and 2,

$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{k[\text{ClO}^-]_2^x [\text{NH}_3]_2^2}{k[\text{ClO}^-]_1^x [\text{NH}_3]_1^2} \text{ where } x \text{ is the order of reaction wrt } \text{ClO}^-$$

$$\frac{2.05}{0.256} = \frac{k(0.4)^x (0.2)^2}{k(0.2)^x (0.1)^2}$$

$$2.001 = \left(\frac{0.4}{0.2}\right)^x \Rightarrow x = 1$$

$$\text{rate} = k[\text{NH}_3]^2[\text{ClO}^-]$$

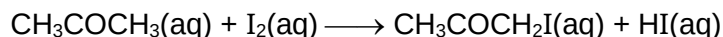
..... [3]

- (iv) Calculate the rate constant, k , for this reaction. Include in your answer, the units of k .

$$k = 128 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$$

..... [2]

- (b) In the presence of an acid catalyst, propanone, CH_3COCH_3 , reacts with iodine according to the following equation.



In a kinetic study, $0.200 \text{ mol dm}^{-3}$ propanone, $0.0100 \text{ mol dm}^{-3}$ iodine and $0.200 \text{ mol dm}^{-3}$ acid was added to a beaker and allowed to react. At fixed time interval, 10 cm^3 samples of the reaction mixture was drawn and analysed. The graph in Fig. 5.1 was obtained.

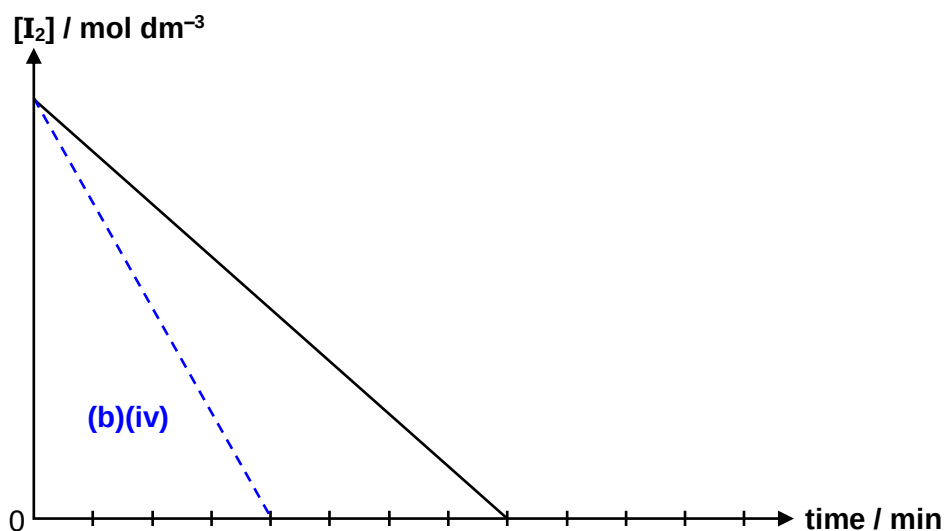


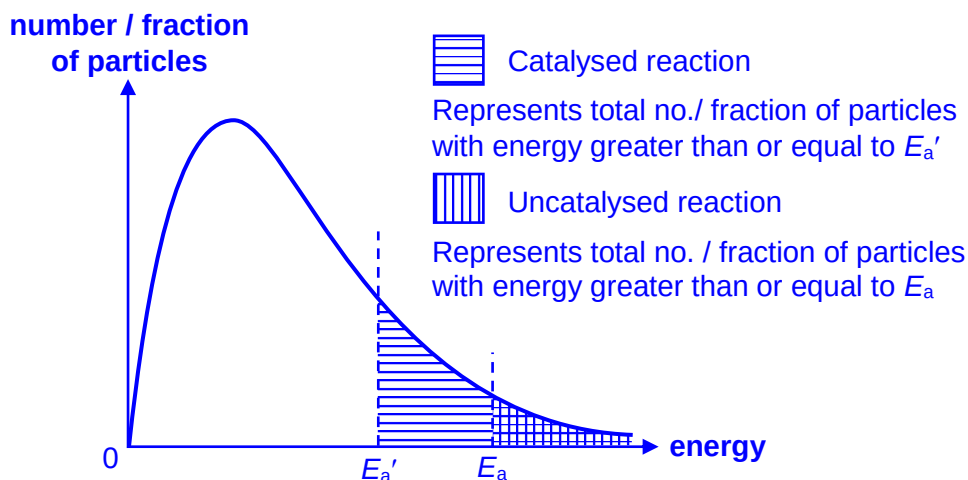
Fig. 5.1

- (i) Suggest how the analysis of the 10 cm^3 samples can be done to monitor rate of reaction.

Monitor the change in $[\text{I}_2]$ by either

- Following the change in colour intensity of iodine since the brown colour fades as reaction progresses. [1]
- or titrating iodine in the 10 cm^3 samples against thiosulfate to determine the $[\text{I}_2]$ left

- (ii) Explain, with a suitable diagram, why the use of a catalyst speeds up the rate of reaction between propanone and iodine.



A catalyst provides an alternative reaction pathway with lower activation energy.

More reactant particles have energy greater than or equal to the new lowered activation energy, E_a' as shown by the diagram.

Hence, frequency of effective collisions increases, rate constant and thus rate increases.

[3]

- (iii) Suggest a reason why the order of reaction with respect to the acid catalyst cannot be deduced in this kinetic study.

A catalyst is regenerated at the end of the experiment.

OR The catalyst is added in large excess.

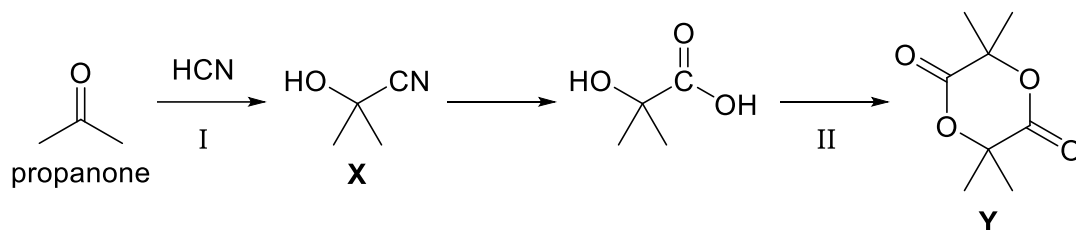
Thus, the concentration of the acid catalyst will remain relatively constant despite the small amount of HI formed, and reaction will appear to be pseudo zero order with respect acid.

[1]

- (iv) Given that the order of reaction with respect to acid is one, sketch on Fig. 5.1, the graph when concentration of acid is doubled while keeping all other concentrations and factors constant.

[1]

- (c) Propanone can also undergo a series of reactions to form carboxylic acid in as shown below.



- (i) Nitrile, -CN , is one of the functional groups present in compound **X**. State the other functional group present in **X**.

Secondary alcohol [1]

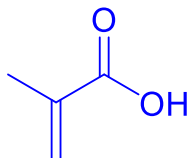
- (ii) State the type of reaction that has occurred in step I.

(nucleophilic) addition [1]

- (iii) State the reagent and condition required for step II.

Concentrated H_2SO_4 , heat under reflux [1]

- (iv) In addition to compound **Y**, in the presence of the reagent and condition stated in (c)(iii), compound **Z**, $\text{C}_4\text{H}_6\text{O}_2$, was formed. Draw the skeletal formula of compound **Z**.



[1]

[Total: 20]

- 6 Sodium is a commercially important alkali metal. As sodium is highly reactive, the elemental metal is stored under oil.

(a) Besides sodium, other group 1 metals also have to be stored carefully to prevent reacting with the atmospheric moisture and oxygen.

(i) State the type of reaction that group 1 metal undergoes with oxygen.

Oxidation..... [1]

(ii) Suggest, with reasoning, the reactivity of group 1 metals with oxygen.

Down the group, the valence electrons are further from the nucleus/ atomic radius becomes larger. With weaker electrostatic attraction between valence electron and nucleus, it becomes easier to lose electron.

..... [2]

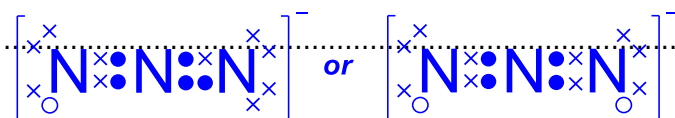
(iii) Group 1 metals, except lithium, do not react readily with nitrogen. Using the *Data Booklet*, explain why only lithium nitride, Li_3N can be formed but sodium nitride, Na_3N cannot be formed.

Compared to Na^+ , the smaller ionic radius of Li^+ with its positive charge contributes to much higher charge density which is compatible with the high charge density of N^{3-} forming strong ionic bond

..... [2]

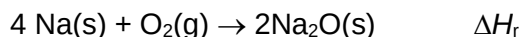
(iv) Group 1 metals, however, can form stable compounds with the azide ion, N_3^- . Azide ion contain two double bonds. Draw the dot and cross diagram of azide ion, stating its shape and bond angle.

Linear about N. 180°



..... [2]

- (b) Sodium burns in oxygen to form sodium oxide, Na_2O . The enthalpy change of reaction between sodium and oxygen is given below.

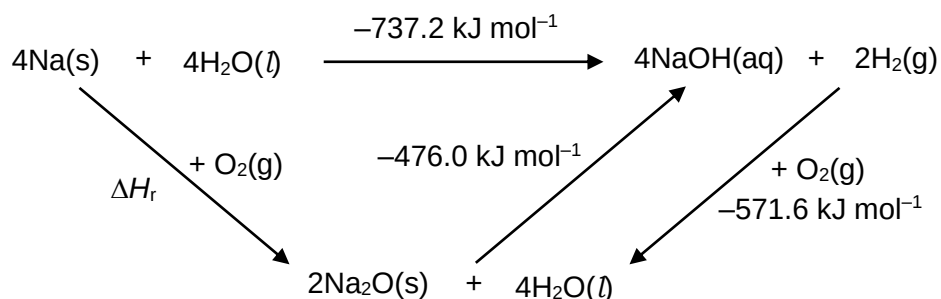


- (i) Define *Hess Law*.

The enthalpy change of a reaction is determined only by the initial and final states and is independent of the reaction pathway taken.

[1]

- (ii) Apply *Hess Law* and calculate the enthalpy change of reaction, ΔH_r , using the energy cycle below.



$$\Delta H_r = -737.2 + (-571.6) - (-476.0)$$

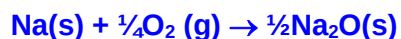
$$= -832.8$$

$$= -833 \text{ kJ mol}^{-1}$$

[2]

- (iii) With the aid of a suitable equation, define the term *enthalpy change of combustion* of sodium.

Enthalpy change of combustion of sodium is the heat evolved when 1 mole of sodium is completely burnt in oxygen.

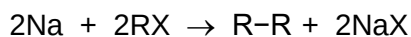


[2]

- (iv) Hence, determine a value for the enthalpy change of combustion of sodium using your answers to (a)(ii) and (iii).

$\Delta H_c = -832.8 \div 4 = -208.2 = -208 \text{ kJ mol}^{-1}$ [1]

- (c) In the Wurtz reaction, sodium reacts with halogenoalkane, RX, to form larger alkane:



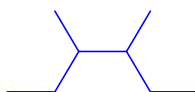
- (i) State the type of reaction that occurred to RX, during the Wurtz reaction.

Substitution..... [1]

- (ii) Name the simplest straight chain alkane that can be formed the Wurtz reaction.

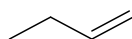
Ethane..... [1]

- (iii) Draw the skeletal formula of corresponding product formed when 2-chlorobutane is used in the Wurtz reaction.

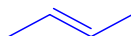


[1]

- (iv) A possible side reaction is the elimination reaction of halogenoalkanes to form alkenes. One such alkene formed from reaction in (c)(iii) is given below:



Draw the other possible alkene that can be formed and suggest, with reasoning, if this other alkene exhibit *cis-trans* isomerism.



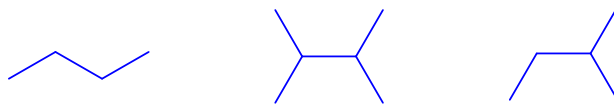
There is *cis-trans* isomerism as there is restricted rotation about the C=C and have different substituents on each C on the C=C.

..... [2]

- (v) In another reaction, two different halogenoalkanes are reacted together with sodium.



Draw the three alkanes that can be formed.



[2]

[Total: 20]

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